Problem 5 – ProofCouncil

A counterexample to the cubic implication for hyperfields

Abstract. In a field, $x+y+z=0$ implies $x^3+y^3+z^3-3xyz=0$. We show that the analogous implication for hyperfields,

$$0 \in x \boxplus y \boxplus z \implies 0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz),$$

fails in general, although it holds in every quotient F/G of a field by a multiplicative subgroup. The counterexample is a hyperfield attached, following Prenowitz and Connes–Consani, to an infinite cyclic projective plane; the plane is built from a perfect difference set in $\mathbb{Z}$ by an explicit variant of Hall's extension argument. All hyperfield axioms are verified directly, so the paper is self-contained.

1 Introduction

Conventions. A *hyperoperation* on a set $\mathbb{H}$ is a map $\boxplus$ assigning to each pair (a,b) a nonempty subset $a \boxplus b \subseteq \mathbb{H}$. It is extended to subsets by the union convention

$$A \boxplus B = \bigcup_{a \in A,\ b \in B} (a \boxplus b), \tag{1}$$

an element being identified with its singleton. Thus $x \boxplus y \boxplus z$ means $(x \boxplus y) \boxplus z = \bigcup_{s \in x \boxplus y} (s \boxplus z)$, and in an expression such as $(-xyz) \boxplus (-xyz) \boxplus (-xyz)$ each occurrence is a separate summand; this is the reading intended in the question below. In a hyperfield (Definition 2.1) the hyperaddition is associative, and then iterated sums do not depend on the bracketing (Remark 2.2).

The question. In a field, $x+y+z=0$ implies $x^3+y^3+z^3-3xyz=0$, because of the polynomial identity

$$x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx). \tag{2}$$

A *hyperfield* is, roughly, a field whose addition is multivalued; the precise definition is recalled in Section 2. The question asks whether the implication survives in that generality.

Question. Let $\mathbb{H}$ be a hyperfield with hyperaddition $\boxplus$, and let $x, y, z \in \mathbb{H}$ satisfy $0 \in x \boxplus y \boxplus z$. Does it follow that

$$0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)\,?$$

Here x^3 is the multiplicative cube and $-xyz$ the additive inverse (the *negative*) of the product xyz. We call the implication asked about the *cubic implication*. The answer is no.

Theorem 1.1. *There exist a countably infinite commutative hyperfield $\mathbb{H}$ and elements $x, y, z \in \mathbb{H}$ such that $0 \in x \boxplus y \boxplus z$ but*

$$0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz).$$

Why a counterexample must be a non-quotient hyperfield. The most familiar hyperfields are Krasner's quotients F/G of a field F by a subgroup G of its multiplicative group [Kr83] (recalled in Example 2.4). In every such hyperfield the cubic implication *does* hold (Proposition 2.5): the hypothesis $0 \in x \boxplus y \boxplus z$ provides representatives $u, v, w \in F$ with $u + v + w = 0$, and the field identity (2) for u, v, w produces the required 0. Consequently a counterexample has to be a hyperfield that is not a quotient of a field, so its hyperfield axioms cannot be inherited from a field; we verify them directly in Section 5, which is the main technical work of the paper.

Strategy. The construction follows the geometric recipe of Prenowitz [Pr43] and Connes–Consani [CC11, §3]. Let P be the point set of a projective plane, in the weak sense that any two distinct points lie on exactly one line and any two distinct lines meet in exactly one point, and write $L(a, b)$ for the line through two distinct points a, b; three points are *collinear* if one line contains them and *noncollinear* otherwise, and the *sides* of a noncollinear triple (a *triangle*) a, b, c are the three lines $L(a, b), L(a, c), L(b, c)$. On $\mathbb{H} = P \cup \{0\}$ define

$$\begin{aligned} a \boxplus b &= L(a, b) \setminus \{a, b\} && (a, b \in P,\ a \neq b), \\ a \boxplus a &= \{0, a\} && (a \in P), \\ 0 \boxplus a &= a \boxplus 0 = \{a\} && (a \in \mathbb{H}) \end{aligned}$$

(this is (6) below). If, in addition, every line is infinite, P is an abelian group, and multiplication by every element of P maps lines to lines, then $\mathbb{H}$ with this hyperaddition and the group multiplication is a commutative hyperfield; this is proved for our plane in Proposition 5.2 by an argument that uses only these properties (compare [CC11, Propositions 3.1 and 3.5], where four points per line suffice). Three features of this hyperfield drive the counterexample.

(a) Every element is its own negative: $0 \in a \boxplus b$ if and only if $a = b$. Hence $-xyz = xyz$, and for subsets $A, B \subseteq \mathbb{H}$ one has $0 \in A \boxplus B$ if and only if $A \cap B \neq \varnothing$ (Lemma 2.3).

(b) If a, b, c are distinct points on a common line L, then $a \boxplus b \boxplus c = L \cup \{0\}$; in particular $0 \in a \boxplus b \boxplus c$.

(c) If a, b, c are noncollinear, then $a \boxplus b \boxplus c = P \setminus \big(L(a,b) \cup L(a,c) \cup L(b,c)\big)$ is the complement of the three sides of the triangle a, b, c (Lemma 5.1); in particular $0 \notin a \boxplus b \boxplus c$.

Now let x, y, z be distinct collinear points, so that $0 \in x \boxplus y \boxplus z$ by (b), and put $q = xyz = -xyz$. The defining rules give $q \boxplus q \boxplus q = \{0, q\} \boxplus q = \{q\} \cup \{0, q\} = \{0, q\}$, so by associativity the six-term sum of the question equals $S \boxplus \{0, q\}$ with $S = x^3 \boxplus y^3 \boxplus z^3$, and by (a) it contains 0 if and only if S meets $\{0, q\}$. Suppose the cubes x^3, y^3, z^3 are noncollinear. Then (c) says that $0 \notin S$ and that $q \in S$ exactly when q lies on none of the sides of the triangle x^3, y^3, z^3. The question has thus become one of pure incidence geometry: the cubic implication fails for x, y, z precisely when the product $q = xyz$ lies on a side of the "cube triangle". So we need a plane whose points form a group acting by collineations, containing three collinear points whose cubes are noncollinear and whose product lies on a side of the cube triangle.

Cyclic planes. Let $P = \{t^j : j \in \mathbb{Z}\}$ be an infinite cyclic group and let $D \subseteq \mathbb{Z}$ be a *perfect difference set*: every nonzero integer has exactly one representation $d - e$ with $d, e \in D$. The sets $L_k = \{t^{k+d} : d \in D\}$, $k \in \mathbb{Z}$, are then the lines of a projective plane on P, and multiplication by t^r carries L_k to L_{k+r} (Section 4). This is the classical correspondence between cyclic planes and difference sets, due to Singer [Si38] for finite planes and to Hall [Ha47] for the group $\mathbb{Z}$. Choose

$$x = 1, \qquad y = t, \qquad z = t^3, \qquad \text{so that} \qquad x^3 = 1, \quad y^3 = t^3, \quad z^3 = t^9, \quad q = xyz = t^4.$$

The requirements become conditions on D: if $0, 1, 3 \in D$ then x, y, z lie on L_0; if $20, 24, 29 \in D$ then 1, $q = t^4$ and t^9 lie on L_{-20}, which is therefore the side $L(1, t^9)$ of the cube triangle; and the uniqueness of differences then forces $23 \notin D$ (because $23 - 20 = 3 - 0$), so that $t^3 \notin L_{-20}$ and the cubes are noncollinear. Hall proved that every finite set of integers with pairwise distinct differences is contained in a perfect difference set of $\mathbb{Z}$ [Ha47, Theorem 3.1]. Section 3 carries this out explicitly, with a concrete spacing rule, for the seed

$$A_0 = \{0, 1, 3, 20, 24, 29\},$$

whose fifteen positive differences are distinct. (The value 20 is one convenient choice making the differences of the seed distinct; nothing else about it is used.)

Organization. Section 2 recalls hyperfields and quotient hyperfields and proves that the cubic implication holds in the latter. Section 3 constructs the perfect difference set D, Section 4 the cyclic plane, and Section 5 the hyperfield $\mathbb{H}$, including the verification of all axioms. Section 6 proves Theorem 1.1 and deduces that $\mathbb{H}$ is not a quotient of a field. Credits for the ingredients are collected in Section 7.

2 Hyperfields and quotient hyperfields

Definition 2.1 (Krasner [Kr83])**.** A *commutative hyperfield* is a set $\mathbb{H}$ with two distinct distinguished elements 0 and 1, a multiplication making $\mathbb{H} \setminus \{0\}$ an abelian group with

identity 1 and satisfying $0 \cdot a = a \cdot 0 = 0$, and a hyperoperation $\boxplus$ (the *hyperaddition*) such that for all $a, b, c \in \mathbb{H}$:

(H1) $a \boxplus b = b \boxplus a$;

(H2) $(a \boxplus b) \boxplus c = a \boxplus (b \boxplus c)$, both sides being read with the convention (1);

(H3) $0 \boxplus a = \{a\}$;

(H4) there is a unique element $-a \in \mathbb{H}$ with $0 \in a \boxplus (-a)$;

(H5) (reversibility) if $c \in a \boxplus b$, then $b \in c \boxplus (-a)$;

(H6) $a(b \boxplus c) = ab \boxplus ac$, where $a(b \boxplus c) = \{ad : d \in b \boxplus c\}$.

A set with a hyperoperation and an element 0 satisfying (H1)–(H5) is called a *canonical hypergroup*, a term due to Mittas [Mi72]; so a hyperfield is a canonical hypergroup with a compatible multiplication. A field is a hyperfield whose sums are singletons. Hyperfields with noncommutative multiplication are also considered in the literature; since the question concerns an arbitrary hyperfield, a commutative counterexample suffices, and "hyperfield" will mean commutative hyperfield below.

Remark 2.2 (iterated sums)**.** The convention (1) makes $\boxplus$ a binary operation on the nonempty subsets of $\mathbb{H}$, and (H1), (H2) make this operation commutative and associative: for nonempty A, B, C,

$$(A \boxplus B) \boxplus C = \bigcup_{a \in A,\, b \in B,\, c \in C} (a \boxplus b) \boxplus c = \bigcup_{a \in A,\, b \in B,\, c \in C} a \boxplus (b \boxplus c) = A \boxplus (B \boxplus C).$$

By the usual generalized associativity for an associative commutative binary operation, an iterated sum $x_1 \boxplus \cdots \boxplus x_n$ is independent of the bracketing and of the order of the summands, and may be regrouped freely. In Section 6 we use this to write a six-term sum as $(x^3 \boxplus y^3 \boxplus z^3) \boxplus (q \boxplus q \boxplus q)$. Note also $A \boxplus \{0\} = A$, by (H3).

Lemma 2.3. *Let $\mathbb{H}$ be a hyperfield and $A, B \subseteq \mathbb{H}$ nonempty. Then $0 \in A \boxplus B$ if and only if there is $a \in A$ with $-a \in B$. In particular, if every element of $\mathbb{H}$ satisfies $-a = a$, then $0 \in A \boxplus B$ if and only if $A \cap B \neq \varnothing$.*

Proof. For elements, $0 \in a \boxplus b$ holds if and only if $b = -a$, by the uniqueness in (H4). Now take the union over $a \in A$, $b \in B$. □

The second statement is the criterion of [CC11, Lemma 3.2], stated there for vector spaces over the Krasner hyperfield $\mathbf{K}$, in which likewise every element is its own negative.

Example 2.4 (quotient hyperfields, Krasner [Kr83])**.** Let F be a field and $G \leq F^\times$ a subgroup of its multiplicative group. Let $F/G = \{0\} \cup \{uG : u \in F^\times\}$ be the set of orbits of G acting on F by multiplication, and let $\pi \colon F \to F/G$ be the orbit map, $\pi(0) = 0$ and

$\pi(u) = uG$ for $u \neq 0$. Multiplication on F/G is defined by $\pi(u)\pi(v) = \pi(uv)$, which is well defined, and hyperaddition by

$$\pi(u) \boxplus \pi(v) = \big\{\pi(u' + v') : \ u' \in \pi^{-1}(\pi(u)),\ v' \in \pi^{-1}(\pi(v))\big\},$$

the set of orbits of all sums of a representative of the first class and a representative of the second. Krasner [Kr83] proved that F/G is a commutative hyperfield. Its negatives are $-\pi(u) = \pi(-u)$, because $0 = \pi(u + (-u)) \in \pi(u) \boxplus \pi(-u)$ and negatives are unique by (H4). The fibres $\pi^{-1}(h)$ are $\{0\}$ and the cosets uG; they are stable under multiplication by elements of G.

Proposition 2.5. *Let F be a field, $G \leq F^\times$ a subgroup, and $x, y, z \in F/G$. If $0 \in x \boxplus y \boxplus z$, then $0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$.*

Proof. We use two general facts about the orbit map π.
(i) *If $u_1, \dots, u_n \in F$, then $\pi(u_1 + \cdots + u_n) \in \pi(u_1) \boxplus \cdots \boxplus \pi(u_n)$.* For $n = 1$ there is nothing to prove. For $n \geq 2$, the element $u_1 + \cdots + u_{n-1}$ is a representative of its own class and u_n of its own, so by the definition of $\boxplus$,

$$\pi(u_1 + \cdots + u_n) \in \pi(u_1 + \cdots + u_{n-1}) \boxplus \pi(u_n),$$

and $\pi(u_1 + \cdots + u_{n-1}) \in \pi(u_1) \boxplus \cdots \boxplus \pi(u_{n-1})$ by induction; the convention (1) combines the two memberships.
(ii) *If $0 \in x \boxplus y \boxplus z$, then there are representatives $u \in \pi^{-1}(x)$, $v \in \pi^{-1}(y)$, $w \in \pi^{-1}(z)$ with $u + v + w = 0$.* By (1), $0 \in s \boxplus z$ for some $s \in x \boxplus y$. By the definition of $\boxplus$, $0 = \pi(s' + w)$ for some $s' \in \pi^{-1}(s)$ and $w \in \pi^{-1}(z)$; as $\pi^{-1}(0) = \{0\}$, this means $s' = -w$. Also $s = \pi(u' + v')$ for some $u' \in \pi^{-1}(x)$, $v' \in \pi^{-1}(y)$. If $u' + v' \neq 0$, then s' and $u' + v'$ lie in the same coset of G, so $s' = (u' + v')g$ for some $g \in G$; put $u = u'g$ and $v = v'g$. These still represent x and y, because fibres are G-stable, and $u + v + w = s' + w = 0$. If $u' + v' = 0$, then $s = 0$, hence $s' = 0$ and $w = 0$, and $u = u'$, $v = v'$ work.

Now let $0 \in x \boxplus y \boxplus z$ and choose u, v, w as in (ii). The identity (2) in the field F gives

$$u^3 + v^3 + w^3 + (-uvw) + (-uvw) + (-uvw) = 0.$$

Since π is multiplicative and $\pi(-a) = -\pi(a)$ (Example 2.4), the six summands represent $\pi(u^3) = x^3$, $\pi(v^3) = y^3$, $\pi(w^3) = z^3$ and $\pi(-uvw) = -xyz$ (three times). Applying (i) to them gives

$$0 = \pi(0) \in x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz). \qquad \square$$

An *isomorphism* of hyperfields is a bijection f preserving 0, multiplication and hyperaddition (that is, $f(a \boxplus b) = f(a) \boxplus f(b)$ as sets); it transports the cubic implication verbatim. Together with Theorem 1.1, Proposition 2.5 will therefore show that the hyperfield constructed below is not isomorphic to any quotient F/G (Corollary 6.2).

3 A perfect difference set in the integers

Definition 3.1. A subset $D \subseteq \mathbb{Z}$ is a *perfect difference set* if the map

$$\{(d,e) \in D \times D : \ d \neq e\} \longrightarrow \mathbb{Z} \setminus \{0\}, \qquad (d,e) \longmapsto d-e,$$

is bijective; that is, every nonzero integer has exactly one expression $d-e$ with $d,e \in D$. For a finite set $A \subseteq \mathbb{Z}$ write

$$\Delta(A) = \{b-a : \ a,b \in A, \ a<b\}$$

for its set of positive differences, and say that A has *distinct differences* if the expressions $b-a$, for $a<b$ in A, are pairwise distinct, i.e. if $|\Delta(A)| = \binom{|A|}{2}$.

The terminology is Hall's [Ha47]; the finite analogue, difference sets modulo n^2+n+1, goes back to Singer [Si38]. Recall from the introduction that, by [Ha47, Theorem 3.1], every finite set with distinct differences is contained in a perfect difference set of $\mathbb{Z}$. We now carry this out for a specific seed and with an explicit spacing rule; the result is Lemma 3.2.

The seed. Let $A_0 = \{0,1,3,20,24,29\}$. Its positive differences, listed by smaller element, are

$$\begin{array}{ll} \text{from } 0: & 1,\ 3,\ 20,\ 24,\ 29, \\ \text{from } 1: & 2,\ 19,\ 23,\ 28, \\ \text{from } 3: & 17,\ 21,\ 26, \\ \text{from } 20: & 4,\ 9, \\ \text{from } 24: & 5, \end{array}$$

so that, in increasing order,

$$\Delta(A_0) = \{1,2,3,4,5,9,17,19,20,21,23,24,26,28,29\}.$$

These are $15 = \binom{6}{2}$ distinct numbers, so A_0 has distinct differences. The roles of the seed elements were explained in the introduction: $0,1,3$ make x,y,z collinear, and $20,24,29$ make $1,t^4,t^9$ collinear.

The extension step. Suppose a finite set $A_n \subseteq \mathbb{Z}_{\geq 0}$ with distinct differences has been constructed. Define

$$M_n = \max A_n, \qquad m_n = \min\big(\mathbb{Z}_{>0} \setminus \Delta(A_n)\big), \qquad N_n = 2M_n + 2m_n + 1, \tag{3}$$

and set

$$A_{n+1} = A_n \cup \{N_n, \ N_n + m_n\}, \qquad D = \bigcup_{n \geq 0} A_n. \tag{4}$$

The integer m_n, the least positive difference not yet realized, exists because A_n is finite. The idea of the step is simple: the two new elements realize the missing difference m_n, and they are placed so far to the right of A_n that every other new difference exceeds every old one. For instance, $m_0 = 6$ and $N_0 = 2 \cdot 29 + 2 \cdot 6 + 1 = 71$, so $A_1 = A_0 \cup \{71, 77\}$; then $m_1 = 7$ and $N_1 = 169$, so $A_2 = A_1 \cup \{169, 176\}$.

Lemma 3.2. *Every A_n has distinct differences. The set D is an infinite perfect difference set with $D \subseteq \mathbb{Z}_{\geq 0}$ and $\min D = 0$.*

Proof. Step 1: A_{n+1} has distinct differences. Abbreviate A_n, M_n, m_n, N_n by A, M, m, N. All elements of A lie in $[0, M]$, so every old difference is at most M. The positive differences of $A_{n+1} = A \cup \{N, N+m\}$ are the old ones, the difference $(N+m) - N = m$ of the two new elements, and the mixed differences

$$N - a \quad \text{and} \quad N + m - a \qquad (a \in A),$$

which are positive because both new elements exceed M. Since $a \leq M$,

$$N - a \geq N - M = M + 2m + 1 > \max\{M, m\},$$

and the same bound holds for $N + m - a$. Consequently no mixed difference coincides with an old difference or with m, and m is not an old difference by its choice. Within the family $\{N - a\}$, equal values force equal a, and likewise within $\{N + m - a\}$. A coincidence $N - a = N + m - b$ between the two families would give $b - a = m$ with $a, b \in A$, i.e. $m \in \Delta(A)$, contradicting the choice of m. Thus all positive differences of A_{n+1} are distinct, and by induction every A_n has distinct differences.

Step 2: every nonzero integer is a difference. By the minimality of m_n, the integers $1, \ldots, m_n - 1$ lie in $\Delta(A_n) \subseteq \Delta(A_{n+1})$, and $m_n \in \Delta(A_{n+1})$ by construction. Hence $\{1, \ldots, m_n\} \subseteq \Delta(A_{n+1})$ and $m_{n+1} > m_n$. The sequence (m_n) is therefore unbounded, so every positive integer k satisfies $k < m_n$ for some n and then $k \in \Delta(A_n)$; thus $k = d - e$ with $d, e \in D$. For negative k, reverse the pair.

Step 3: uniqueness. Suppose $k = d - e = d' - e'$ with $(d, e) \neq (d', e')$ in $D \times D$ and, say, $k > 0$. The sets A_n increase with n, so d, e, d', e' all lie in one A_n, and k would have two different expressions as a positive difference in A_n, contrary to Step 1. For $k < 0$ negate both expressions.

Step 4. Each step adjoins two elements larger than M_n, so D is infinite; and $0 \in A_0 \subseteq D \subseteq \mathbb{Z}_{\geq 0}$. □

4 The cyclic plane

Let

$$P = \{t^j : \ j \in \mathbb{Z}\}$$

be an infinite cyclic group with generator t, written multiplicatively: $t^i t^j = t^{i+j}$, with identity $1 = t^0$, and distinct exponents denote distinct elements. Using the perfect difference set D of Lemma 3.2, define for every $k \in \mathbb{Z}$ the *line*

$$L_k = \{t^{k+d} : \ d \in D\} = t^k L_0.$$

Thus the lines are the translates of the single line $L_0 = \{t^d : d \in D\}$. Three points are *collinear* if some line contains all of them, and *noncollinear* otherwise; a noncollinear triple

a, b, c is also called a *triangle* with *vertices* a, b, c, and once (I1) below is available its *sides* are the lines $L(a, b), L(a, c), L(b, c)$. The following lemma collects everything that will be used about the plane.

Lemma 4.1. *The points of P and the lines L_k have the following properties.*

(I1) Any two distinct points of P lie on exactly one line.

(I2) Any two distinct lines meet in exactly one point.

(I3) The lines L_k, $k \in \mathbb{Z}$, are pairwise distinct.

(I4) Every line is infinite. (Also, no line is all of P, since $t^{k-1} \notin L_k$; this will not be needed.)

(I5) $t^r L_k = L_{k+r}$ for all $r, k \in \mathbb{Z}$.

Proof. (I3) Since $\min D = 0$, the least exponent occurring in L_k is k.

(I1) Let $t^i \neq t^j$, so $i \neq j$. The point t^i lies on L_k if and only if $i - k \in D$. Hence L_k contains both points if and only if $i - k = d$ and $j - k = e$ for some $d, e \in D$, and then $d - e = i - j \neq 0$. By Lemma 3.2 there is exactly one pair $(d, e) \in D \times D$ with $d - e = i - j$, so exactly one k, namely $k = i - d = j - e$, works; by (I3) this means that exactly one line contains both points.

(I2) Let $k \neq \ell$. A point t^j lies on both L_k and L_ℓ if and only if $j - k = d \in D$ and $j - \ell = e \in D$, and then $d - e = \ell - k \neq 0$. Again there is exactly one such pair (d, e), hence exactly one such $j = k + d$.

(I4) D is infinite, and $-1 \notin D$.

(I5) $t^r L_k = \{t^{r+k+d} : d \in D\} = L_{k+r}$. □

By (I1) we may write $L(a, b)$ for the unique line through two distinct points a, b; thus, for $c \neq a, b$, the points a, b, c are collinear if and only if $c \in L(a, b)$. A useful consequence of (I5): for distinct $a, b \in P$ and $u \in P$,

$$uL(a, b) = L(ua, ub), \tag{5}$$

because $uL(a, b)$ is a line by (I5) and contains the distinct points ua, ub. Properties (I1) and (I2) say that P with the lines L_k is a projective plane in the weak sense used in the introduction, and (I5) says that the group P acts on it by collineations. Projective planes admitting a cyclic group of collineations that acts sharply transitively on the points (for any two points exactly one group element carries the first to the second) are the *cyclic planes* of Singer and Hall; in the terminology of [CC11, §3], a group structure on the point set for which multiplications are collineations, such as our P, is a two-sided incidence group of the plane.

5 The hyperfield

Adjoin to P a new element $0 \notin P$ and put $\mathbb{H} = P \cup \{0\}$. Extend the multiplication of P by $0a = a0 = 0$ for all $a \in \mathbb{H}$; the multiplicative identity is $1 = t^0$, which is different from the new element 0. Define the hyperaddition by

$$\begin{aligned} 0 \boxplus a = a \boxplus 0 &= \{a\} && (a \in \mathbb{H}), \\ a \boxplus a &= \{0, a\} && (a \in P), \\ a \boxplus b &= L(a,b) \setminus \{a, b\} && (a, b \in P,\ a \neq b). \end{aligned} \tag{6}$$

In words: the sum of two distinct points consists of the other points of the line joining them, and the sum of a point with itself is the point together with 0. This is Prenowitz's hyperaddition on a projective geometry [Pr43], in the form used by Connes and Consani [CC11, Proposition 3.1], who show that it yields a canonical hypergroup whenever every line has at least four points, and that a two-sided incidence group turns it into a hyperfield [CC11, Proposition 3.5]. We do not rely on those results but verify the axioms directly; this also makes visible exactly which incidence properties are used, namely (I1), (I2), (I4) and (5).

Throughout this section we write, for $a, b, c \in \mathbb{H}$,

$$T(a,b,c) = (a \boxplus b) \boxplus c = \bigcup_{d \in a \boxplus b} (d \boxplus c)$$

for the left-bracketed triple sum. The first lemma is the heart of the associativity proof: for a triangle the triple sum has a description that is visibly symmetric in the three summands. The formula also appears in the proof of [CC11, Proposition 3.1].

Lemma 5.1 (triangle formula)**.** *For three noncollinear points $a, b, c \in P$,*

$$T(a,b,c) = (a \boxplus b) \boxplus c = P \setminus \big(L(a,b) \cup L(a,c) \cup L(b,c)\big). \tag{7}$$

In particular $0 \notin T(a,b,c)$, and $T(a,b,c)$ is disjoint from the three sides of the triangle abc.

Proof. Noncollinear points are distinct, and $c \notin L(a,b)$, $b \notin L(a,c)$, $a \notin L(b,c)$; so the three sides are three distinct lines, and by (I2) any two of them meet only in their common vertex.

The left side is contained in the right side. Let $r \in T(a,b,c)$. Then $r \in d \boxplus c$ for some $d \in a \boxplus b = L(a,b) \setminus \{a,b\}$. Since $c \notin L(a,b)$ we have $d \neq c$, so $d \boxplus c = L(d,c) \setminus \{d,c\}$; in particular $r \neq 0$, $r \in P$, and $r \neq d, c$. The line $L(d,c)$ is different from $L(a,b)$ (it contains c), so it meets $L(a,b)$ only in d, by (I2); as $r \neq d$, $r \notin L(a,b)$. Next, $d \notin L(a,c)$: the lines $L(a,c)$ and $L(a,b)$ are distinct and meet only in a, while $d \in L(a,b)$ and $d \neq a$. Hence $L(d,c) \neq L(a,c)$, these two lines meet only in c, and $r \neq c$ gives $r \notin L(a,c)$. The same argument with b in place of a gives $r \notin L(b,c)$.

The right side is contained in the left side. Let $r \in P$ lie on none of the three sides. Then $r \neq c$, and the line $L(c,r)$ is different from $L(a,b)$ because $c \notin L(a,b)$; by (I2) the two lines meet in exactly one point d. If d were equal to a, then $a \in L(c,r)$ would force

$L(c,r) = L(a,c)$ by (I1), placing r on the side $L(a,c)$; so $d \neq a$, and likewise $d \neq b$. Also $d \neq c$ (as $c \notin L(a,b)$) and $d \neq r$ (as $r \notin L(a,b)$). Therefore $d \in L(a,b) \setminus \{a,b\} = a \boxplus b$, and $r \in L(c,r) \setminus \{d,c\} = L(d,c) \setminus \{d,c\} = d \boxplus c$, so $r \in T(a,b,c)$. □

Proposition 5.2. *With the multiplication and hyperaddition above, $\mathbb{H}$ is a commutative hyperfield in the sense of Definition 2.1. Moreover*

$$0 \in a \boxplus b \quad \Longleftrightarrow \quad a = b \qquad (a, b \in \mathbb{H}), \tag{8}$$

so every element is its own negative, $-a = a$.

Proof. All sums are nonempty: a line minus two points is nonempty by (I4). Commutativity (H1) and the neutrality of 0 (H3) are built into (6).

Negatives (H4). Inspecting the three rules of (6) gives (8): 0 lies in $a \boxplus a$ for every a (including $a = 0$), in $0 \boxplus b = \{b\}$ only if $b = 0$, and in $L(a,b) \setminus \{a,b\} \subseteq P$ never. Thus a itself is the unique element $-a$ with $0 \in a \boxplus (-a)$.

Reversibility (H5). Since $-a = a$, we must show that $c \in a \boxplus b$ implies $b \in c \boxplus a$; we show in fact that the two conditions

$$c \in a \boxplus b \qquad \text{and} \qquad b \in c \boxplus a$$

are equivalent. If one of a, b, c is 0, both conditions say that the other two are equal: for $a = 0$ by (H3), for $b = 0$ by (H3) and (8), and for $c = 0$ by (8) and (H3). Let $a, b, c \in P$. If $a = b = c$, both conditions hold ($a \in a \boxplus a$). If exactly two of them are equal, both conditions fail: a sum $u \boxplus u = \{0, u\}$ never contains a point different from u, and a sum $u \boxplus v$ with $u \neq v$ contains neither u nor v. If a, b, c are distinct, both conditions say that a, b, c are collinear, by (I1).

Associativity (H2). The commutativity part of Remark 2.2 uses only (H1), so

$$a \boxplus (b \boxplus c) = (b \boxplus c) \boxplus a = T(b, c, a).$$

Hence (H2) follows once we show that $T(a,b,c)$ depends only on the multiset $\{a,b,c\}$ and not on the order of the summands. A triple either contains 0, or lies in P^3 and has three equal summands, exactly two equal summands, or three distinct summands, the last being collinear or not; we compute T in each of these five cases.

- *A zero summand.* By (H3) and $A \boxplus \{0\} = A$, $T(0,a,b) = T(a,0,b) = T(a,b,0) = a \boxplus b$, which is symmetric in a, b.
- *Three equal nonzero summands a, a, a.* $T(a,a,a) = \{0,a\} \boxplus a = (0 \boxplus a) \cup (a \boxplus a) = \{0, a\}$.
- *Exactly two equal summands.* Let the summands be a, a, b with $a \neq b$ in P, in any order, and put $L = L(a,b)$. Then

$$T(a,a,b) = \{0,a\} \boxplus b = \{b\} \cup \big(L \setminus \{a,b\}\big) = L \setminus \{a\},$$

and, since $d \boxplus a = L \setminus \{d, a\}$ for $d \in L \setminus \{a, b\}$,

$$T(a, b, a) = T(b, a, a) = \bigcup_{d \in L \setminus \{a,b\}} \big(L \setminus \{a, d\}\big) = L \setminus \{a\}.$$

For the last equality: the union is contained in $L \setminus \{a\}$, and a point $r \in L \setminus \{a\}$ is retained by choosing $d \in L \setminus \{a, b, r\}$, which is possible because L is infinite (I4). So all three orderings give $L(a, b) \setminus \{a\}$.

- *Three distinct collinear summands* a, b, c on a line L. Here $a \boxplus b = L \setminus \{a, b\}$, and for $d \in L \setminus \{a, b\}$ the summand $d \boxplus c$ is $\{0, c\}$ if $d = c$ and $L \setminus \{d, c\}$ if $d \neq c$ (because then $L(d, c) = L$). Hence

$$T(a, b, c) = \{0, c\} \cup \bigcup_{d \in L \setminus \{a,b,c\}} \big(L \setminus \{d, c\}\big) = \{0, c\} \cup \big(L \setminus \{c\}\big) = L \cup \{0\},$$

where a point $r \in L \setminus \{c\}$ is retained by choosing $d \in L \setminus \{a, b, c, r\}$, again possible by (I4). The result $L \cup \{0\}$ depends only on the line L, hence only on the multiset $\{a, b, c\}$.

- *Three noncollinear summands.* By Lemma 5.1, $T(a, b, c)$ is the complement in P of the union of the three sides, which is symmetric in a, b, c.

This proves (H2). Thus $(\mathbb{H}, \boxplus, 0)$ is a canonical hypergroup. (It also satisfies $a \boxplus \mathbb{H} = \mathbb{H}$, a condition some authors include: given b, pick $c \in b \boxplus a$; then $b \in c \boxplus a$ by (H5).)

Multiplication and distributivity (H6). The nonzero elements form the abelian group P with identity $1 \neq 0$, and 0 is absorbing. Let $u \in P$. For distinct $a, b \in P$, (5) gives

$$u(a \boxplus b) = uL(a, b) \setminus \{ua, ub\} = L(ua, ub) \setminus \{ua, ub\} = ua \boxplus ub,$$

using that multiplication by u is a bijection of P. For equal nonzero summands, $u\{0, a\} = \{0, ua\} = ua \boxplus ua$; if a summand is 0, both sides of (H6) are the singleton of the product of u with the other summand. Finally, for $u = 0$ both sides of (H6) equal $\{0\}$. □

Remark 5.3 (where the size of the lines enters)**.** Infiniteness of the lines was used only to choose a point of a line outside a prescribed set of at most four points; so the proof works verbatim for any plane satisfying (I1), (I2), whose point set is an abelian group acting by collineations as in (I5), and in which every line has at least five points. Connes and Consani show that four points per line suffice for the hypergroup axioms [CC11, Proposition 3.1]; on a four-point line $L = \{a, b, c, d\}$ one computes $a \boxplus b \boxplus c = \{0, a, b, c\}$ rather than $L \cup \{0\}$, still a symmetric expression. With only three points per line the rule (6) is not associative: on the line $\{a, b, c\}$ one finds $(a \boxplus b) \boxplus c = c \boxplus c = \{0, c\}$ but $a \boxplus (b \boxplus c) = a \boxplus a = \{0, a\}$.

6 The counterexample

We now prove Theorem 1.1 for the hyperfield $\mathbb{H}$ of Proposition 5.2 and the elements

$$x = 1, \qquad y = t, \qquad z = t^3.$$

Recall that $D \supseteq A_0 = \{0, 1, 3, 20, 24, 29\}$ and that $-a = a$ for every $a \in \mathbb{H}$, so the summands of the conclusion are

$$x^3 = 1, \qquad y^3 = t^3, \qquad z^3 = t^9, \qquad -xyz = xyz = t^4 =: q.$$

Theorem 6.1. *In $\mathbb{H}$, with $x = 1$, $y = t$, $z = t^3$,*

$$0 \in x \boxplus y \boxplus z \qquad \textit{but} \qquad 0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz).$$

Proof. Step 1: the hypothesis holds. Since $0, 1, 3 \in D$, the three distinct points $1, t, t^3$ lie on L_0. In particular $t^3 \in L(1, t) \setminus \{1, t\} = 1 \boxplus t$, hence

$$0 \in t^3 \boxplus t^3 \subseteq (1 \boxplus t) \boxplus t^3 = x \boxplus y \boxplus z. \tag{9}$$

(In fact $x \boxplus y \boxplus z = L_0 \cup \{0\}$ by the collinear case in the proof of Proposition 5.2, but (9) is all we need.)

Step 2: the cubes are noncollinear, and q lies on a side of their triangle. Since $20, 24, 29 \in D$,

$$1 = t^{-20+20}, \quad q = t^4 = t^{-20+24}, \quad t^9 = t^{-20+29} \ \in \ L_{-20}, \tag{10}$$

so $L_{-20} = L(1, t^9)$ by (I1). On the other hand $23 \notin D$: otherwise the two distinct pairs $(23, 20)$ and $(3, 0)$ in $D \times D$ would both represent $3 = 23 - 20 = 3 - 0$, contradicting the uniqueness in Lemma 3.2. Hence $t^3 = t^{-20+23} \notin L_{-20} = L(1, t^9)$, which by (I1) says precisely that $1, t^3, t^9$ are noncollinear.

Step 3: the conclusion fails. Put

$$S = x^3 \boxplus y^3 \boxplus z^3 = (1 \boxplus t^3) \boxplus t^9 = T(1, t^3, t^9).$$

By Step 2 and the triangle formula (Lemma 5.1), S contains neither 0 nor any point of the side $L(1, t^9) = L_{-20}$. By (10), $q \in L_{-20}$, so

$$S \cap \{0, q\} = \varnothing. \tag{11}$$

Next, by (6),

$$q \boxplus q \boxplus q = \{0, q\} \boxplus q = (0 \boxplus q) \cup (q \boxplus q) = \{q\} \cup \{0, q\} = \{0, q\}.$$

By Remark 2.2 the six-term sum may be regrouped:

$$x^3 \boxplus y^3 \boxplus z^3 \boxplus q \boxplus q \boxplus q = (x^3 \boxplus y^3 \boxplus z^3) \boxplus (q \boxplus q \boxplus q) = S \boxplus \{0, q\}.$$

Finally, Lemma 2.3 together with $-a = a$ gives, for subsets $A, B \subseteq \mathbb{H}$, that $0 \in A \boxplus B$ if and only if $A \cap B \neq \varnothing$. Applied to $A = S$ and $B = \{0, q\}$, (11) yields

$$0 \notin S \boxplus \{0, q\} = x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz).$$

Together with (9) this proves the theorem. □

Theorem 6.1 establishes Theorem 1.1: $\mathbb{H}$ is countably infinite, since $P \cong \mathbb{Z}$, and commutative by Proposition 5.2. So the answer to the question is **no**.

Corollary 6.2. *The hyperfield $\mathbb{H}$ is not isomorphic to a quotient hyperfield F/G of a field F by a subgroup $G \leq F^\times$.*

Proof. By Proposition 2.5 every quotient hyperfield satisfies the cubic implication, an isomorphism of hyperfields transports the implication, and $\mathbb{H}$ violates it by Theorem 6.1. □

Remark 6.3. Hyperfields that are not quotients of fields were first constructed by Massouros [Ma85], by a different method. Hyperfields attached to infinite cyclic planes built from Hall's difference sets appear in [CC11, Example 4.7]. For quotient hyperfields F/G satisfying $a \boxplus a = \{0, a\}$ for all a (a property our $\mathbb{H}$ shares, by (6)), the projective geometry attached to the hyperaddition, whose lines are the sets $\{a, b\} \cup (a \boxplus b)$ for distinct nonzero a, b (for our $\mathbb{H}$ these are exactly the lines $L(a, b)$), is Desarguesian; [KB25, Example 2.6(4) and Remark 4.3(3)] contrasts such quotients with a hyperfield built on an infinite cyclic group whose plane is non-Desarguesian, and thereby separates general hyperfields from quotients of fields. Corollary 6.2 gives, for the specific $\mathbb{H}$ constructed here, a direct elementary verification that involves no Desargues configuration: the cubic implication is a first-order statement true in every F/G and false in $\mathbb{H}$.

Remark 6.4 (what the example shows)**.** For distinct $x, y, z \in P$ the hypothesis $0 \in x \boxplus y \boxplus z$ is the collinearity of x, y, z (by the collinear and noncollinear cases in the proof of Proposition 5.2), and, since cubing is injective on $P \cong \mathbb{Z}$ so that the cubes are distinct, the conclusion holds if the cubes are collinear and otherwise holds exactly when xyz avoids all three sides of the cube triangle, by the argument of Step 3. In a field, or in a quotient of a field, the two conditions are linked by the identity (2). In the cyclic plane of a perfect difference set the two conditions are independent, and the construction exploits this: the seed elements $0, 1, 3 \in D$ make x, y, z collinear, so the hypothesis holds; the seed elements $20, 24, 29 \in D$ place $xyz = t^4$ on the side $L(1, t^9)$ of the cube triangle, so the conclusion fails; and the uniqueness of differences built into D excludes $23 \in D$ (which would duplicate the difference $3 = 23 - 20 = 3 - 0$) and so keeps the cubes noncollinear.

7 Background and credits

None of the ingredients is new: the hyperfield axioms and the quotient hyperfields F/G are Krasner's [Kr83], the term canonical hypergroup is Mittas's [Mi72], the hyperaddition "third points of the joining line" is Prenowitz's [Pr43] and was combined with an incidence group to produce hyperfields by Connes and Consani [CC11, §3, Example 4.7], and perfect difference sets and cyclic planes are due to Singer [Si38] (finite case) and Hall [Ha47] (the group $\mathbb{Z}$), of whose Theorem 3.1 Lemma 3.2 is an explicit instance. Proposition 2.5 is an elementary observation about quotient hyperfields. What appears to be new is the counterexample itself: the choice of the seed A_0, the resulting incidence configuration, and the failure of the cubic implication; a literature search did not locate the question or an earlier answer.

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